TASK 2 CASE STUDY: DRUG FORMULATION

Formulation of a Novel Transdermal Patch

Utilizing Siegesbeckiae Herba Extract for the Alternative Therapy of Rheumatoid Arthritis

PRESENTER: CODEALPHA PHARMACEUTICAL INTERN

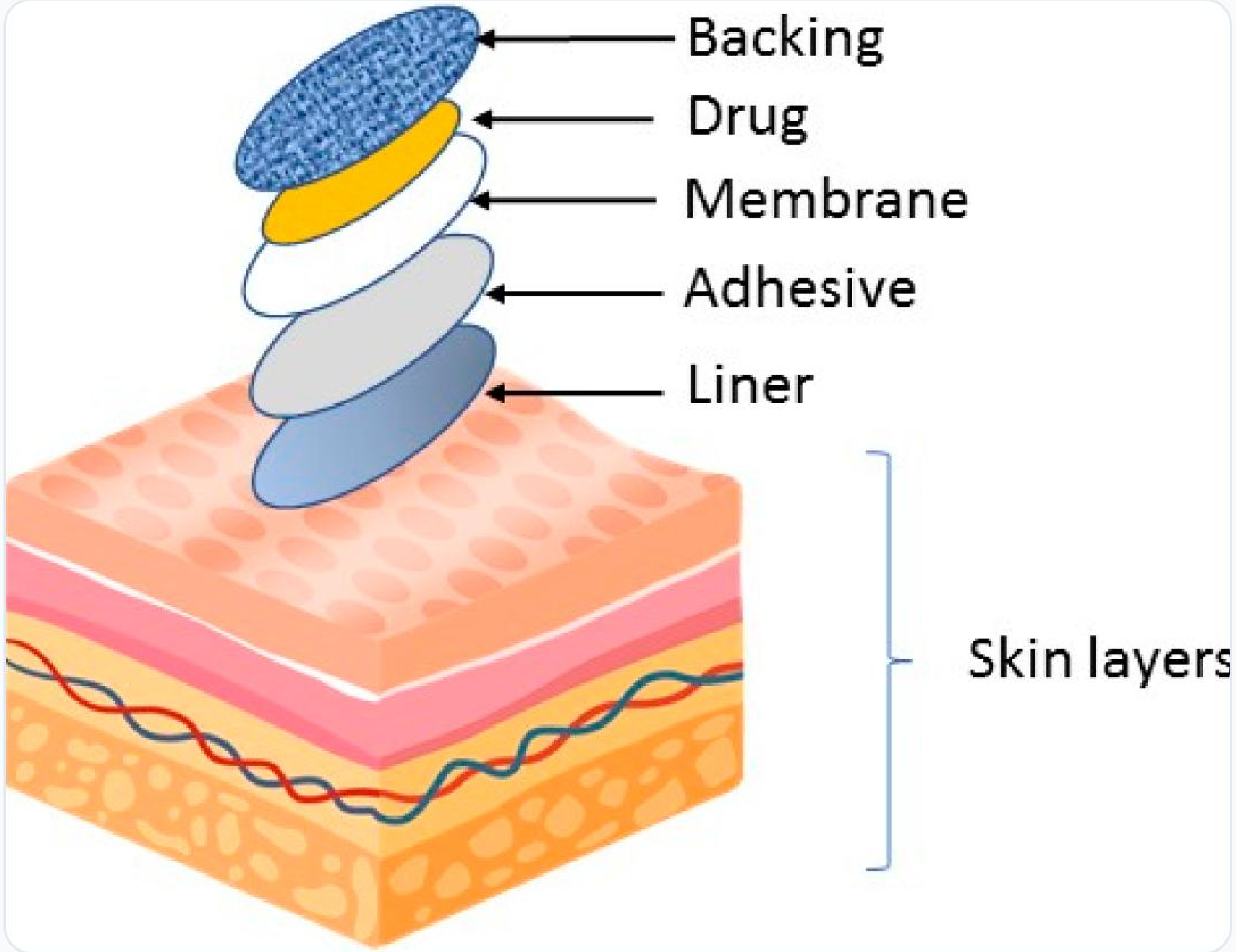
Dosage Form Selection & Rationale

Targeting Rheumatoid Arthritis (RA)

Rheumatoid arthritis is a chronic autoimmune disease characterized by progressive joint destruction and severe localized pain, affecting approximately 1% of the global population.

Why a Transdermal Patch?

- **Avoids Oral Risks:** Bypasses oral NSAID pathways, reducing gastrointestinal bleeding and cardiovascular complications.
- **Sustained Delivery:** Maintains continuous, controlled plasma drug concentrations.
- **Patient Compliance:** Offers a non-invasive, painless, self-administered therapeutic solution.



Patch Preparation Process

The transdermal drug-in-adhesive (DIA) patches were manufactured using a controlled **solvent evaporation technique**, ensuring highly uniform active ingredient dispersion.

1. Active Extraction

Siegesbeckiae Herba processed twice with 70% ethanol under sonication; filtered and vacuum dried into a 48.5% kirenol-rich powder.

2. Matrix Blending

SiH extract, oleic acid (penetration enhancer), and DURO-TAK 87-2287 PSA polymer dissolved in ethanol and agitated for 2 hours.

3. Precision Coating

Coated onto a Scotchpak™ 9744 release liner using a wet film applicator, set at room temp, then oven-dried at 50°C for 10 min.

4. Lamination & Cut

Laminated with a Scotchpak™ 9701 backing film barrier, cooled, and precisely cut to predefined active treatment dimensions.

Formulation Components Matrix

Detailed structure of the optimized pressure-sensitive adhesive (PSA) transdermal delivery matrix:

Component Class	Material Selected	Critical Functional Role	Proportion (w/w)
Active Pharmaceutical Ingredient	Siegesbeckiae Herba (SiH) Extract	Diterpenoid active (Kirenol) providing anti-inflammatory & analgesic efficacy.	10.0% (0.1 g)
Permeation Enhancer	Oleic Acid (OA)	Disrupts stratum corneum lipids; breaks PSA-drug hydrogen bonds.	10.0% (0.1 g)
Pressure Sensitive Adhesive	DURO-TAK® 87-2287	Hydroxyl-functional acrylic polymer matrix hosting the active system.	80.0% (1.0 g)
Backing Membrane	Scotchpak™ 9701	Provides outer occlusive support, stopping volatile active loss.	Physical Layer
Release Liner	Scotchpak™ 9744	Peelable fluoropolymer protective release liner layer.	Physical Layer

Permeation Enhancer Efficacy

To bypass the stratum corneum barrier, multiple enhancers were screened. **Oleic Acid (OA)** yielded the highest cumulative kirenol skin penetration:

> *Mechanistic Note: OA interacts with the intercellular lipid bilayers of the stratum corneum, fluidizing the lipids. Its carboxyl group breaks key hydrogen bonds between the adhesive's hydroxyl functional groups and kirenol to trigger rapid drug diffusion.*

Critical Stability Challenges

Challenge 1: Drug Loading Threshold

While transdermal drug delivery scales with the concentration gradient, active ingredients cannot uniformly disperse at high loads. Increasing SiH extract to 15% (w/w) caused phase separation and recrystallization. The drug load was stabilized strictly at 10% (w/w) to preserve patch uniformity.

Challenge 2: Adhesive Hydrogen Bonding

Adhesives containing carboxyl groups (such as DURO-TAK 87-2852) interacted too strongly with kirenol, locking the active inside. Hydroxyl-functional DURO-TAK 87-2287 was mandatory to ensure optimal cohesive properties and highly

Pharmacodynamic Evaluation

77.88%

Pain Inhibition Ratio (PIR)

Virtually identical to commercial Diclofenac patches (78.13%), proving outstanding analgesic alternative potential.

Formulation Evaluation Formulas

1. Paw Swelling Degree (S %)

$$S\,\% = \frac{V_t - V_n}{V_n} \times 100$$

2. Pain Inhibition Ratio (PIR %)

$$PIR\,\% = \frac{W_e - W_t}{W_e} \times 100$$

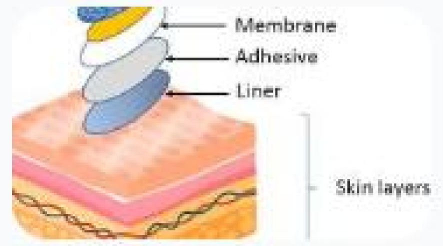
Questions & Answers

This completed Task 2 Case Study details the selection, manufacturing process, excipient composition, optimization, and clinical validation of a modern transdermal botanical patch.

Thank you for your attention!

Source Data Reference: Journal of Ethnopharmacology 265 (2021) 113294
"Alternative therapy of rheumatoid arthritis with a novel transdermal patch containing Siegesbeckiae Herba extract"

Image Sources



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